



Decision Trees

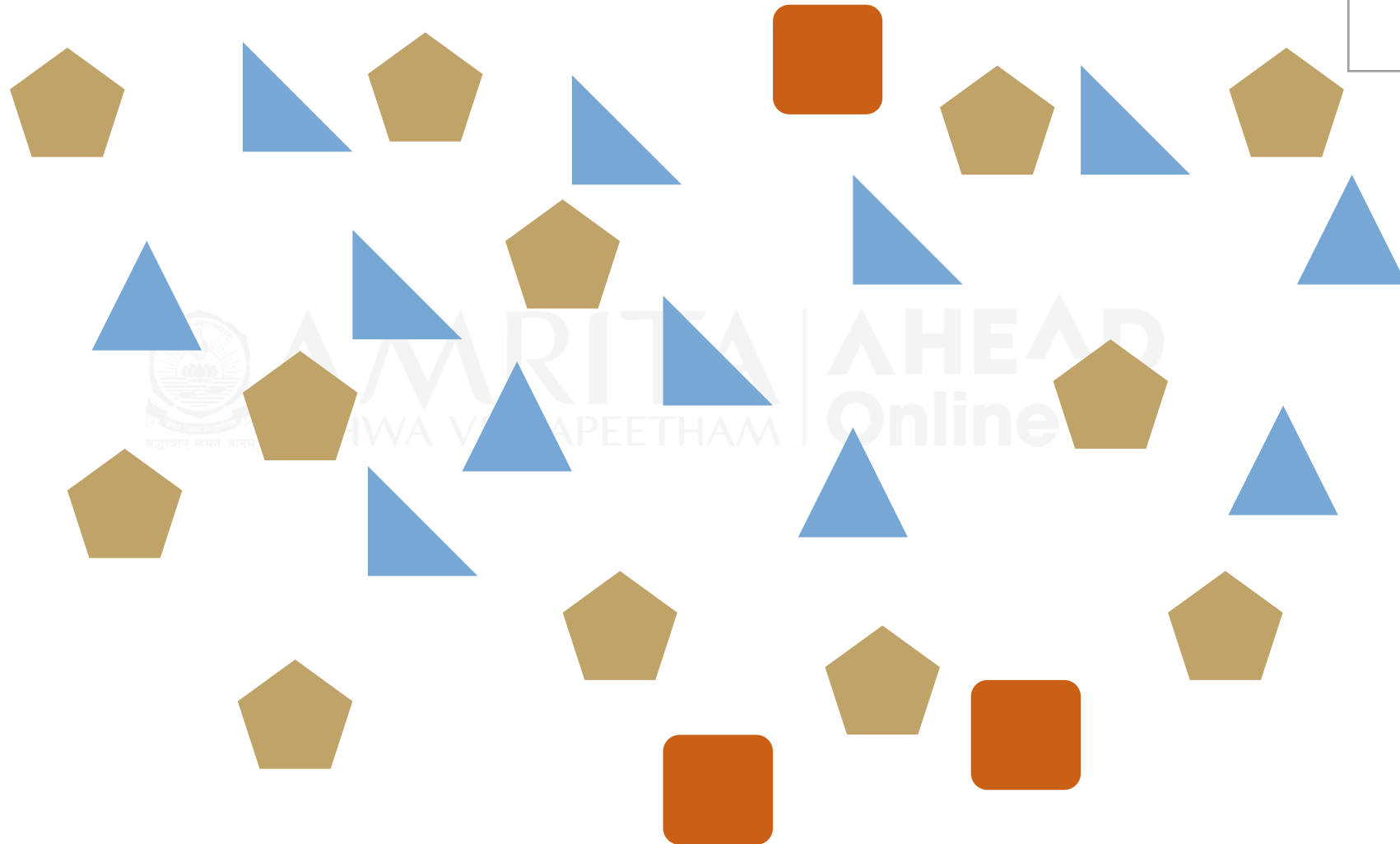
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Amrita School of Engineering

Introduction

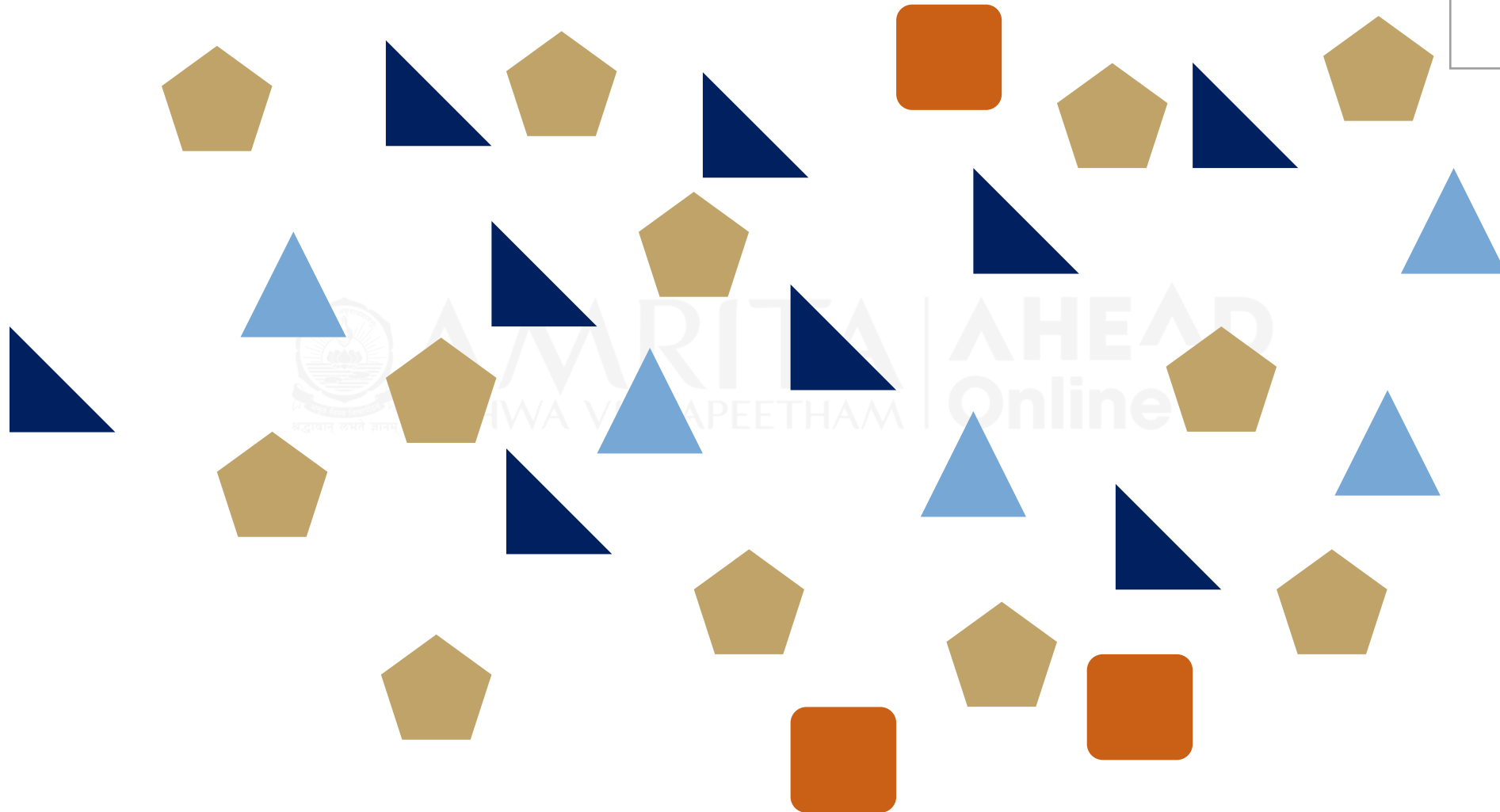
- Finding rules to split up triangles
- Terminology for a tree
- Finding a good attribute to split
- Spam Tree

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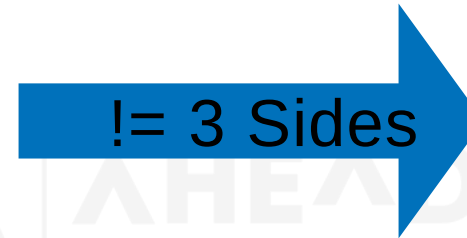
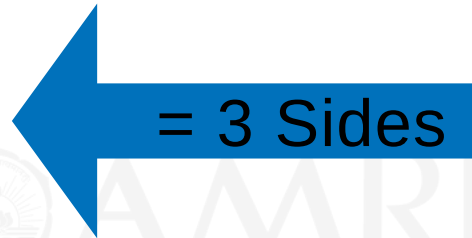
Find the Triangles



Find the Triangles



Find the Triangles

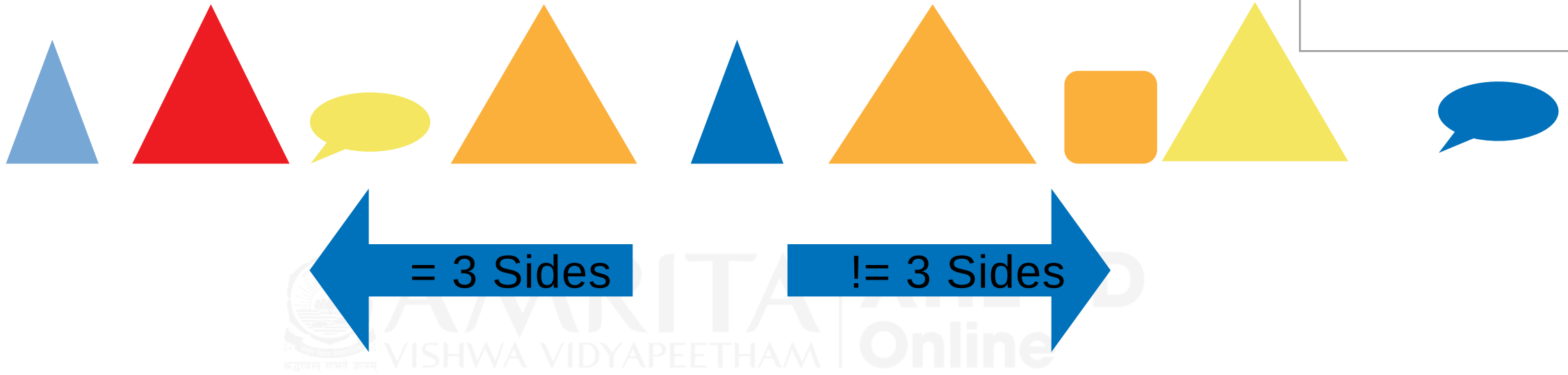


AMRITA
VISHWA VIDYAPEETHAM

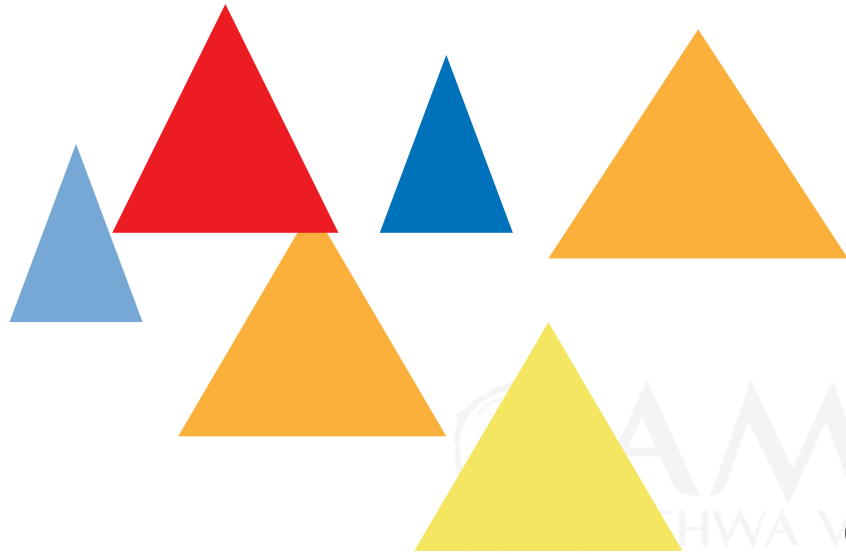
AHEAD
Online



Find the Orange Triangles Step1

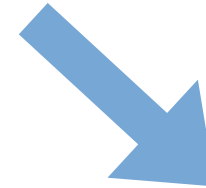
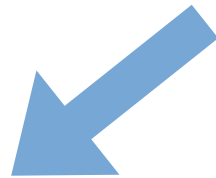


Find the Orange Triangles Step2

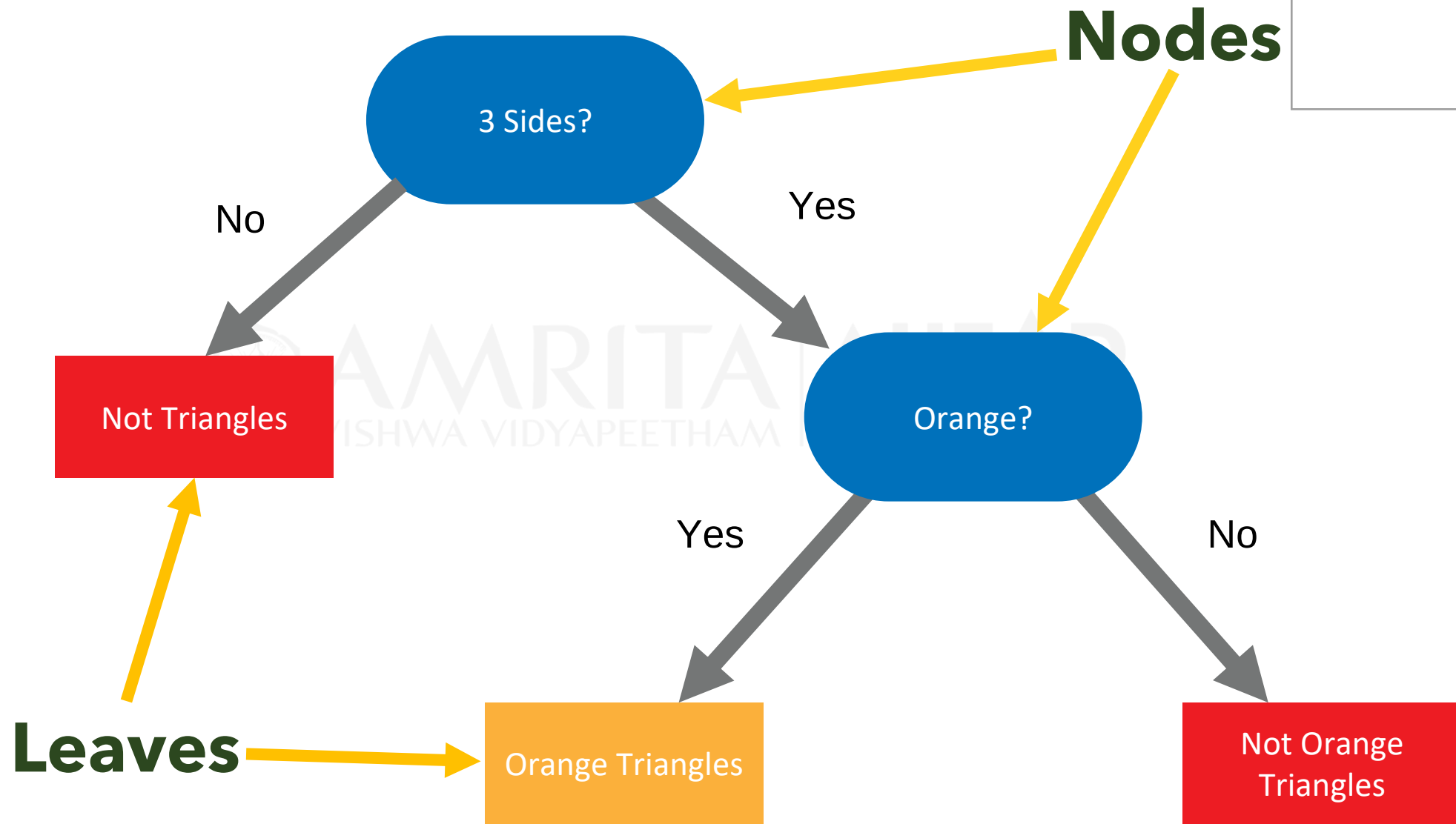


Orange

NOT Orange



Tree for Orange Triangles



Find the best attribute to separate the data



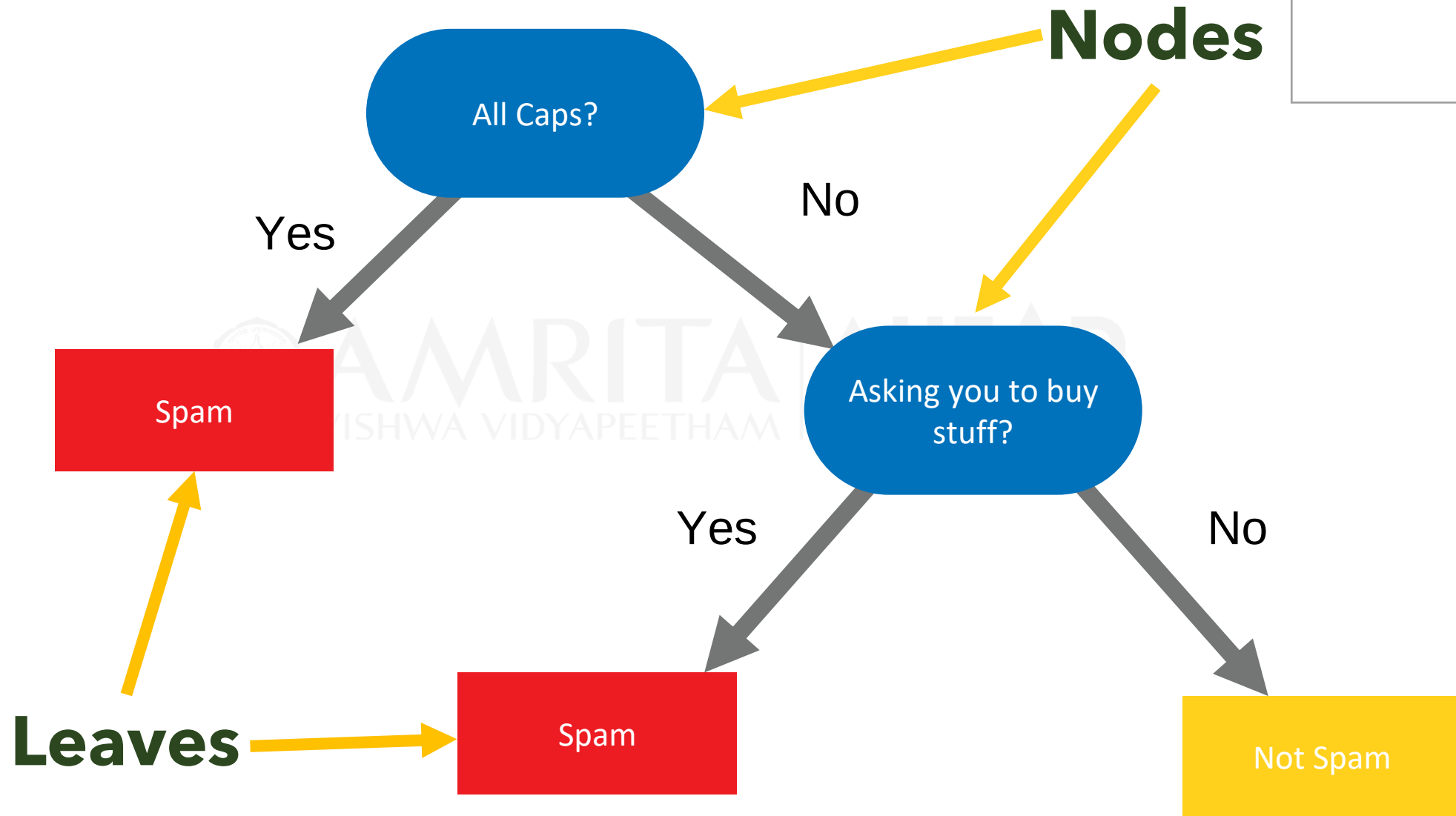
Spam	Email
yes	Congratulations Claim one of our free exclusive reward offers (minimum value \$90).
no	Dear John, I didn't see you today. Are you feeling well?
no	Hey, Yesterday I forgot to turn off the lights, can you please check the room for me?
yes	<u>TELL US WHERE TO SEND YOUR PACKAGE! 100% All Natural Pure Ketosis Formula CONFIRM YOUR EXCLUSIVE TRIAL NOW! if you can't see the images below please , click here</u>

Best Attributes

Spam	Email
yes	Congratulations Claim one of our free exclusive reward offers (minimum value \$90).
no	Dear John, I didn't see you today. Are you feeling well?
no	Hey, Yesterday I forgot to turn off the lights, can you please check the room for me?
yes	TELL US WHERE TO SEND YOUR PACKAGE! 100% All Natural Pure Ketosis Formula CONFIRM YOUR EXCLUSIVE TRIAL NOW! if you can't see the images below please , click here

- ALL CAPS
- Grammatical or not
- Claim / Reward / Get
- Links / no links

Tree for Spam



Conclusion

- Decision tree is a bunch of if-else statements
- We want to split up the data until we “know” what the class is
- Certain features will be better at splitting than others

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Decision Trees II

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Introduction

- Intuitions for a tree
- Pseudocode
- Base cases
- Tree properties

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Intuitions for A Decision Tree



- Using relevant attributes construct a series of if-else statements
- You can do this repeatedly until you have a tree that separates the data
- Decision trees are constructed with greedy algorithms - they always go for the “best possible” split, recursively



PseudoCode (Wikipedia)



- Check for base cases.
- For each feature **A**, calculate a splitting “value” (how good will it be to split on this value)
- Let **A_best** be the attribute with the highest splitting “value”
- Create a decision node that splits on **A_best**.
- Recur on the sublists obtained by splitting on **A_best**, and add those nodes as children of node.

Base Cases



- All the samples belong to the same class. When this happens, it simply creates a leaf node for the decision tree saying to choose that class.
- None of the features provide any "value". You can't find a good split - some heuristic strategy must be used.

Properties Of Decision Trees



Pros

- Training and Inference is Fast
- Memory space required is small
- Can handle categorical data without modification
- Can handle multi-class classification without modification
- Scaling data is not required
- White box models - Interpretability
- We can “look and see” why a decision was made

Cons

- Trend towards very high complexity (overfitting)

Conclusion

- Learned about decision trees
- Greedy algorithms that always overfit
- Next step is to figure out how to formalize a good split

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Splitting Criterion

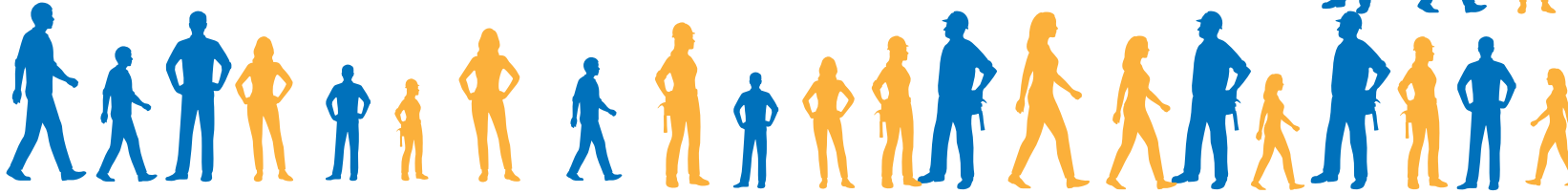
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Introduction

- What makes a good basketball player?
- What makes a good split
- Gini impurity coefficient
- Information gain ratio
- It's all about purity

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WHO PLAYS BASKETBALL?



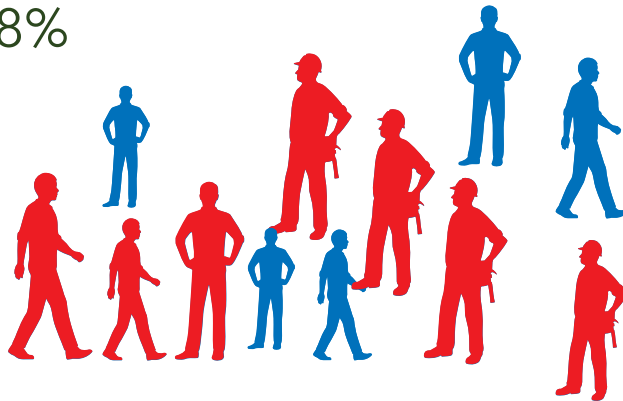
24 People

12/24 or 50% Play Basketball

Split on Gender

People : 12 Play Basketball: 7

$7/12 = 58\%$



People: 12, Play Basketball: 5

$5/12 = 42\%$



WHO PLAYS BASKETBALL?

24 People



12/24 or 50% Play Basketball

People : 14 Play Basketball:
10

$10/14 = 71.4\%$



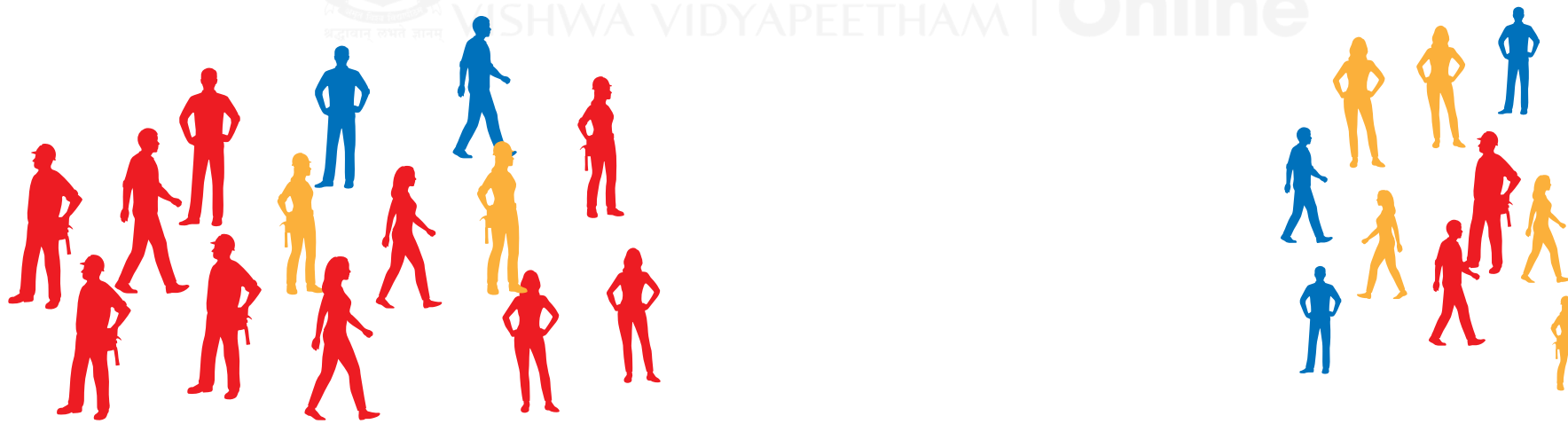
Split on Height
> 5'11" ft

People: 10, Play Basketball: 2
 $2/10 = 20\%$



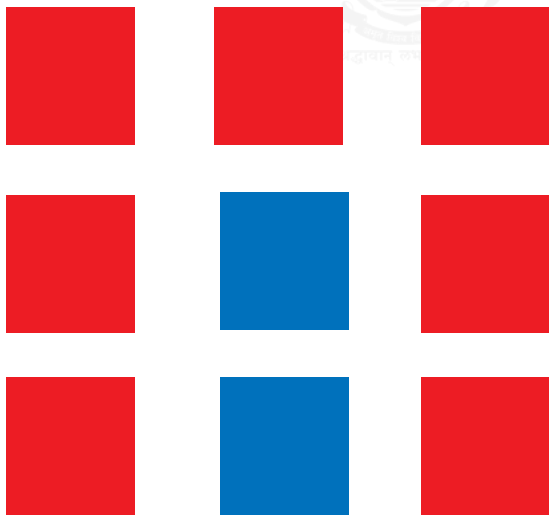
Good Splits Maximize “purity”

- We want to split on features that will separate our data into “pure” groups.
- We can quantify this with math



Gini Impurity Coefficient

- “Gini impurity is a measure of how often a randomly chosen element from the set would be incorrectly labeled if it was randomly labeled according to the distribution of labels in the subset.”



- $1 - \sum_{i=1}^j p_i^2$

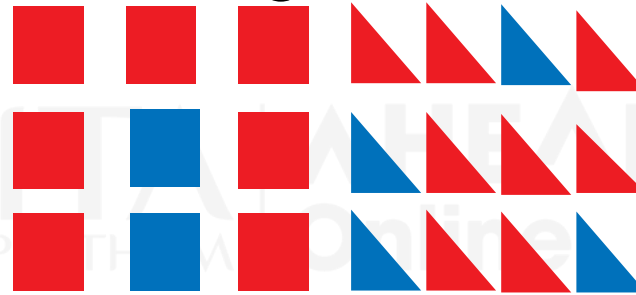
- $1 - (2/9)^2 - (7/9)^2$

- 0.345

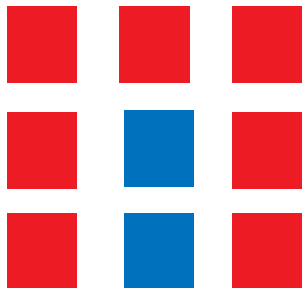
https://en.wikipedia.org/wiki/Decision_tree_learning

Gini Impurity Coefficient

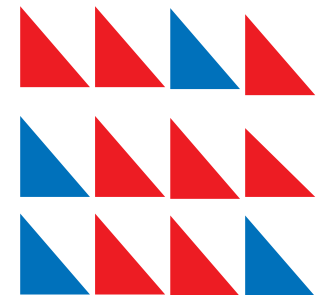
- We need to calculate the Gini impurity for **both** nodes created from the split. Then we take a weighted average and this is our final number



- $1 - (2/9)^2 - (7/9)^2$
- 0.345



- $1 - (4/12)^2 - (8/12)^2$
- .444



- Weighted Average
- $.345 * 9/21 + .444 * 12/21$
- .402**

Information gain Ratio



- Entropy - how much uncertainty is there in a coin toss?
 - Maximum entropy (1 bit)
- We want to reduce entropy as much as possible:
 - Entropy of parent - sum(entropy children)
- This is *information gain ratio* - how much information did we gain by doing the split?

It's all about Purity



- Both information gain ratio and gini measure *impurity* in the sub-nodes.
- Decision trees will iterate over all features looking for the split that maximizes purity.
- Whatever is most pure is used - this is a greedy algorithm.

Conclusion

- We want our split to create pure sub-groups
- Mathematically maximizing purity is our aim

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Lab Solution Decision Tree Spam

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Introduction

- Investigating a dataset
- Classifying spam w/
Decision Trees
- Fighting overfitting
- Tuning the hyper-parameters

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Conclusion

- Dataset investigation
- Observing overfitting in default trees
- Hyperparameter overfitting
 - Overfit to the test data when tuning hyper-parameters.
- Brainstorm ideas?

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